```
Start coding or generate with AI.
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Name: Aman Shaikh **Div**: F **Roll No**: 02

Task

Perform K-Means clustering on the 'Annual Income (k$)' and 'Spending Score (1-100)' features from the "Mall_Customers.csv" dataset. Determine the optimal number of clusters using the Elbow Method and visualize the WCSS against K. Then, apply K-Means with the optimal K, visualize the clustered data with distinct colors for each cluster and marked cluster centers, and summarize the insights gained from the clustering.

Load Dataset

```
import pandas as pd
df = pd.read_csv('/content/Mall_Customers.csv')
df.head()
```

	CustomerID	Genre	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

Select Features

Subtask:

Select 'Annual Income (k$)' and 'Spending Score (1-100)' as the features for K-Means clustering.

```
X = df[['Annual Income (k$)', 'Spending Score (1-100)']]
X.head()
```

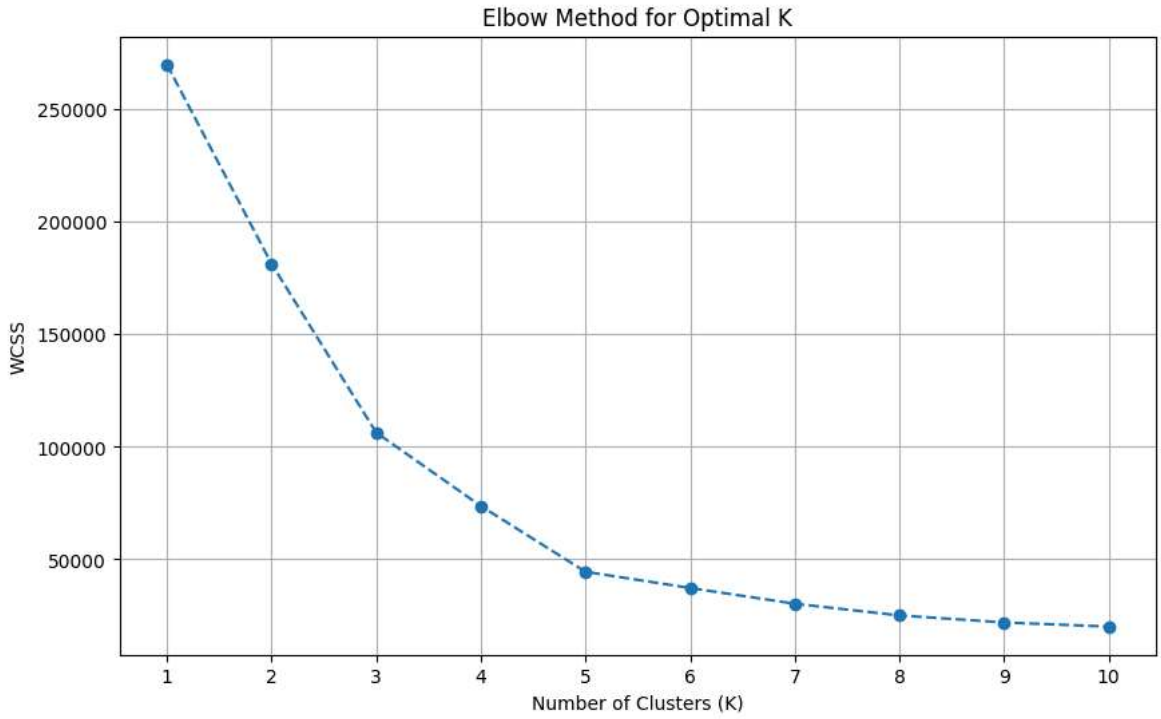
	Annual Income (k$)	Spending Score (1-100)
0	15	39
1	15	81
2	16	6
3	16	77
4	17	40

Determine Optimal K (Elbow Method)

```
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

wcss = []
for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, init='k-means++', random_state=42, n_init=10)
    kmeans.fit(X)
    wcss.append(kmeans.inertia_)

plt.figure(figsize=(10, 6))
plt.plot(range(1, 11), wcss, marker='o', linestyle='--')
plt.title('Elbow Method for Optimal K')
plt.xlabel('Number of Clusters (K)')
plt.ylabel('WCSS')
plt.xticks(range(1, 11))
plt.grid(True)
plt.show()
```



Apply K-Means Clustering

```
optimal_k = 5
kmeans = KMeans(n_clusters=optimal_k, init='k-means++', random_state=42, n_init=10)
kmeans.fit(X)
X['Cluster'] = kmeans.labels_
centers = kmeans.cluster_centers_

print("Cluster Labels:")
print(X['Cluster'].value_counts())
print("\nCluster Centers:")
print(centers)
```

```
Cluster Labels:
Cluster
0    81
1    39
3    35
4    23
2    22
Name: count, dtype: int64

Cluster Centers:
[[55.2962963  49.51851852]
 [86.53846154 82.12820513]
 [25.72727273 79.36363636]
 [88.2        17.11428571]
 [26.30434783 20.91304348]]
/tmp/ipython-input-958346181.py:4: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view
  X['Cluster'] = kmeans.labels_
```

```
optimal_k = 5
kmeans = KMeans(n_clusters=optimal_k, init='k-means++', random_state=42, n_init=10)
kmeans.fit(X)
X.loc[:, 'Cluster'] = kmeans.labels_
centers = kmeans.cluster_centers_

print("Cluster Labels:")
print(X['Cluster'].value_counts())
print("\nCluster Centers:")
print(centers)
```

```
Cluster Labels:
Cluster
0    81
1    39
3    35
```

```
4    23
2    22
Name: count, dtype: int64

Cluster Centers:
[[ 5.52962963e+01  4.95185185e+01 -1.99840144e-15]
 [ 8.65384615e+01  8.21282051e+01  1.00000000e+00]
 [ 2.57272727e+01  7.93636364e+01  2.00000000e+00]
 [ 8.82000000e+01  1.71142857e+01  3.00000000e+00]
 [ 2.63043478e+01  2.09130435e+01  4.00000000e+00]]
```

```
X = df[['Annual Income (k$)', 'Spending Score (1-100)']].copy()

optimal_k = 5
kmeans = KMeans(n_clusters=optimal_k, init='k-means++', random_state=42, n_init=10)
kmeans.fit(X)
X.loc[:, 'Cluster'] = kmeans.labels_
centers = kmeans.cluster_centers_

print("Cluster Labels:")
print(X['Cluster'].value_counts())
print("\nCluster Centers:")
print(centers)
```

```
Cluster Labels:
Cluster
0    81
1    39
3    35
4    23
2    22
Name: count, dtype: int64

Cluster Centers:
[[55.2962963  49.51851852]
 [86.53846154 82.12820513]
 [25.72727273 79.36363636]
 [88.2        17.11428571]
 [26.30434783 20.91304348]]
```

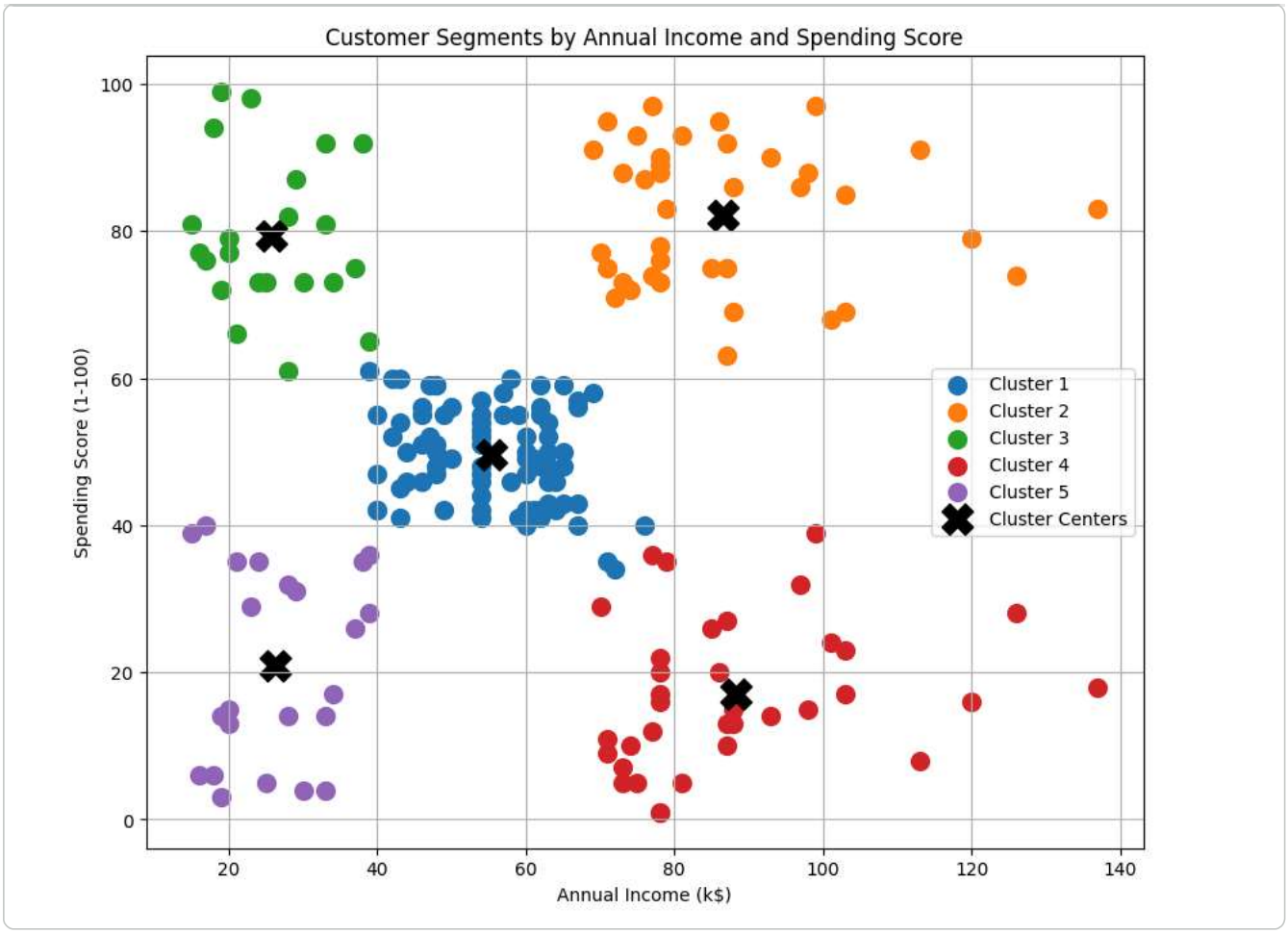
Visualize Clustered Data

```
plt.figure(figsize=(10, 8))

# Plotting the clusters
for i in range(optimal_k):
    plt.scatter(
        X[X['Cluster'] == i]['Annual Income (k$)'],
        X[X['Cluster'] == i]['Spending Score (1-100)'],
        s=100, label=f'Cluster {i+1}'
    )

# Plotting the cluster centers
plt.scatter(
    centers[:, 0],
    centers[:, 1],
    s=300, c='black', marker='X', label='Cluster Centers'
)

plt.title('Customer Segments by Annual Income and Spending Score')
plt.xlabel('Annual Income (k$)')
plt.ylabel('Spending Score (1-100)')
plt.legend()
plt.grid(True)
plt.show()
```


Customer Segments by Annual Income and Spending Score
Spending Score (1-100)
Annual Income (k$)
0
20
40
60
80
100
20
40
60
80
100
120
140
Cluster 1
Cluster 2
Cluster 3
Cluster 4
Cluster 5
Cluster Centers